Displaying Descriptive Statistics

Dr. Skip Crooker MBA 775 University of Nevada, Las Vegas

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Objectives

- Read a chart and say what it does — and does not — support.
- Tell *discrete* data from *continuous* data, and say why it changes the chart — and, later, the probability distribution.
- Construct and interpret a *frequency distribution* and *histogram*, grouped and ungrouped.
- Construct and interpret *ogives* and *cumulative* frequencies.
- Display qualitative data using *bar charts* and *pie charts*.
- Display quality issues with a *Pareto chart*.
- Organize two variables in a *contingency table* and read a *heatmap*.
- Present a distribution with a *stem and leaf* display.
- Show the relationship between two variables using a *scatter plot*, and say which variable belongs on which axis.
- Recognize the ways a technically correct chart can still mislead.
- Notice when two variables are not in comparable *units*, and say what the comparison is actually measuring.
- Specify a chart or a frequency distribution precisely enough that an AI assistant can build it in Excel — and verify what comes back.

The code that draws each chart is folded away. Click **Show the code** if you are curious, but nothing in this chapter is assessed on your ability to write it. What is assessed is whether you can look at the output and say what it means.

The role technology plays in statistics

Hardcore statisticians are perfectly happy to receive large quantities of data in tabular formats. Likely this is because they know how to import the data into their preferred statistical interface and can perform their own analysis on it. Artificial intelligence tools are cutting the gap between what requires a hardcore statistician and what an interested person can do.

Senior leaders often have the analytical ability to draw insight from raw data. What they do not have is the time. Even traditional descriptive statistics — **mean**, **median**, **quartiles**, **minimum** and **maximum** — take more effort to interpret than many will spend. That is a particular challenge when your goal is to *persuade*. We do not want our audience's eyes to glaze over. We want to make the case from the data as elegantly as possible.

This brings us to **data visualization**: presenting data so that its important features are clear. This takes skill. There are many chart types to choose from, and some are far better than others at showing a particular pattern. AI is a genuine boon to producing them — and no help at all in deciding which one is honest.

Seeing a pattern that a table hides

Two centuries of data, as a table

Here is the Maddison Project's estimate of GDP per capita, for every country it covers, for every year it covers.

```
21,586 rows
178 countries and regions
years 1 to 2022

     entity code  year  gdp_pc
Afghanistan  AFG  1950  1156.0
Afghanistan  AFG  1951  1170.0
Afghanistan  AFG  1952  1189.0
Afghanistan  AFG  1953  1240.0
Afghanistan  AFG  1954  1245.0
Afghanistan  AFG  1955  1246.0
Afghanistan  AFG  1956  1278.0
Afghanistan  AFG  1957  1253.0
Afghanistan  AFG  1958  1298.0
Afghanistan  AFG  1959  1307.0
```

Ten of twenty-one thousand rows. Nobody is going to read this, and printing more of it would not help. So narrow it all the way down — one row per observation for the *world*, with the share of people living in extreme poverty beside it.

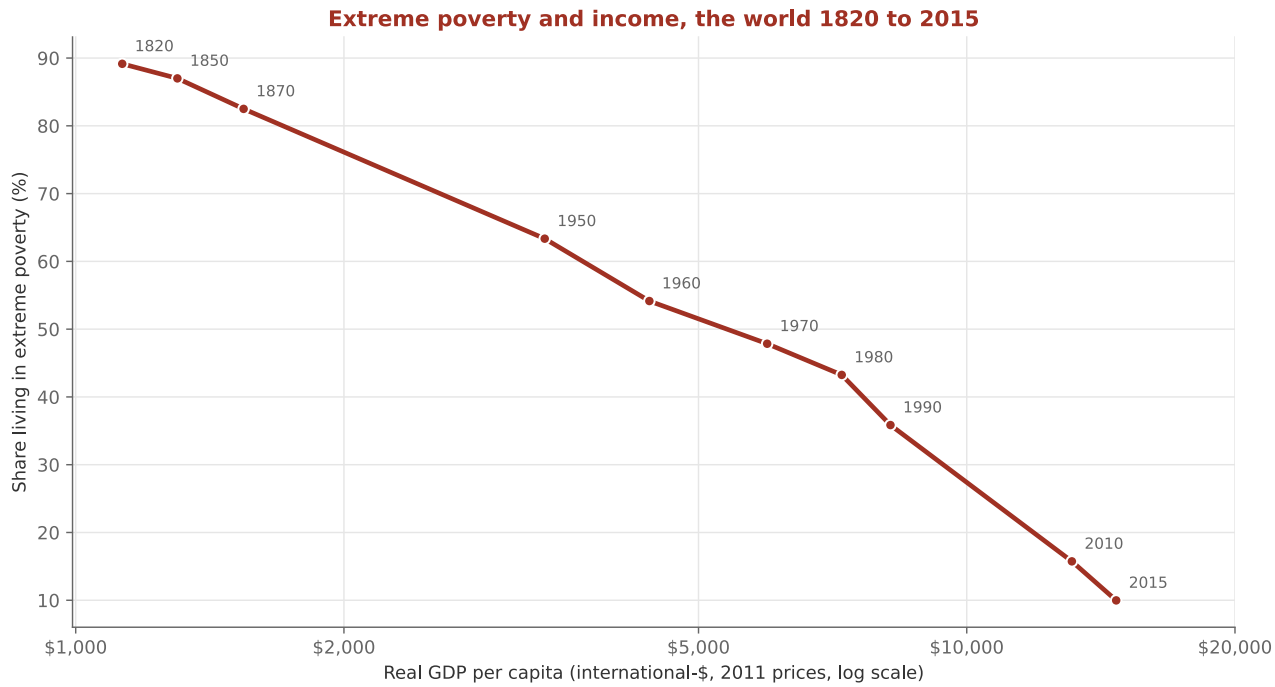
Year	GDP per capita	% in extreme poverty	In poverty (millions)	World population (millions)
1820	1,127.7	89.2	964.9	1,082.4
1850	1,300.5	87.0	1,098.5	1,262.7
1870	1,543.2	82.5	1,154.4	1,399.3
1950	3,360.2	63.4	1,596.4	2,520.0
1960	4,403.7	54.1	1,638.6	3,026.0
1970	5,970.7	47.9	1,766.2	3,691.2
1980	7,239.2	43.2	1,920.1	4,439.6
1990	8,211.0	35.9	1,903.5	5,309.7
2010	13,121.6	15.7	1,090.7	6,929.7
2015	14,713.5	10.0	733.5	7,349.5

Ten rows. This one you can read, and reading it tells you two things immediately: income went up, and the share in poverty went down.

Now say what *shape* that relationship has. Is it steady? Does it accelerate? Does it flatten out at high incomes? Is it tight, or is it loose with a lot of exceptions?

You cannot answer any of that from the table, and the table is only ten rows long. This is the part worth being precise about: **the table is not hiding the facts. It is hiding the structure.**

The same ten rows, drawn



Neither axis is time. Both variables are measured quantities, and the points are joined in the order they happened — so the line is the path the world has travelled through two centuries. It goes one way.

```
1820    89.2% in extreme poverty   GDP per capita $1,128   world population 1.08 billion
2015    10.0% in extreme poverty   GDP per capita $14,713   world population 7.35 billion
```

> In 1820, **nine people in ten** on this planet lived in extreme poverty. By 2015, **one in ten** did.
>
> There is no precedent for that in recorded history, and no table of ten rows makes you feel it.

How strong is it, and why the axis is logarithmic

```
correlation of poverty share with GDP per capita       r = -0.976
correlation of poverty share with LOG GDP per capita   r = -0.990
```

Both are strong. The second is stronger, and that is the justification for the log axis rather than a decorative preference: income spans more than an order of magnitude over this period — from about $1,100 to about $15,000 — and the relationship is closer to straight when income is expressed in *multiples* rather than in dollars.

> This is a real choice, and it belongs in the “charts that mislead” discussion later in this chapter as much as here. On a linear axis the same ten points bunch up at the left and the last century dominates the picture. On a log axis they spread evenly and the whole two hundred years is legible.
>
> Neither version is false. They answer different questions, and the axis is where you decide which question you are asking.

The number that did not fall

Now the harder half of the same data, from the same two columns.

```
Number of people in extreme poverty
  1820       965 million
  1980     1,920 million   <- the peak
  2015       733 million

The SHARE fell from 89.2% to 10.0%.
The COUNT fell by only 24%.
World population grew 6.8 times.
```

The number of people living in extreme poverty **peaked in 1980**, at nearly two billion — more than twice the 1820 figure — during a period when the *share* was falling the whole time. And in 2015 it was still 733 million people.

> Both of these are true, computed from the same two columns:
>
> Extreme poverty has almost been eliminated — it fell from 89% to 10%.
>
> Seven hundred and thirty-three million people still live in extreme poverty, barely fewer than in 1820.
>
> Which one you put in the caption decides what your reader believes, and neither one is a lie. This is the Valentine's Day pie chart from later in this chapter, at world-historical scale: **the arithmetic is not the argument. The choice of denominator is.**

What this chart is, and what it is not

Four things belong in the provenance record before anyone quotes a number off it.

```
Poverty : World population living in extreme poverty
          Ravallion (2016), updated with World Bank (2019), via
          Our World in Data
Line    : $1.90 per day, international-$ at 2011 prices
Income  : Maddison Project Database 2023 (Bolt and van Zanden),
          constant international-$ at 2011 prices, via Our World in Data
Coverage: 1820 to 2015, 10 observations
```

> **What "constant international-$ at 2011 prices" actually says.** Two separate adjustments are buried in that phrase, and neither is the one most people assume.
>
> - **Constant** means adjusted across *time* — for inflation. So this is **real** GDP per capita, not nominal.
> - **International-$** means adjusted across *countries* — for purchasing power parity, so that a dollar buys a comparable basket everywhere.
>
> The base is **2011**, which is not a recent year. Note the source: Maddison Project Database *2023*. That 2023 is the year the database was **released**, not the year its prices are measured in. Reading $14,713 for 2015 as though it were in today's dollars would be off by a decade and a half of inflation.
>
> So the 1820 figure reads: *the average person's annual output that year, valued at what $1,128 would have bought in the United States in 2011.* Not 1820 dollars. Not today's dollars.
>
> **And here is why that matters for this particular chart.** The poverty line is *also* measured at 2011 prices. Both variables share a base year. Had the income series been in 2011 dollars and the poverty line in 2021 dollars, putting them on one chart would have been the same mistake you are about to watch a house price index make — two variables that look comparable and are not.

That the bases match is not luck. It is the thing to check before drawing.

1. **The poverty line here is superseded.** These figures use $1.90 a day at 2011 prices. The World Bank has since moved to $2.15 at 2017 prices and then to $3.00 at 2021 prices. Each revision changes how many people are counted as extremely poor without anyone's circumstances changing at all — the same lesson as the two consumer confidence series later in this chapter, applied to the most-quoted statistic in development economics.
2. **The early numbers are reconstructions, not measurements.** Nobody surveyed household consumption in 1820. The figure is modelled from national accounts and distribution estimates, and different teams have published different reconstructions — some put 1820 nearer three-quarters than nine-tenths. **The level is uncertain. The direction is not.**
3. **It stops in 2015**, because that is where this particular series stops.
4. **It is a time path, not a comparison of countries.** The claim it supports is about the world getting richer and less poor together over two hundred years. It is *not* the claim that a rich country today has less poverty than a poor one — that would need a different chart, drawn from the country panel rather than the world row.

With all four caveats stated, the finding survives every one of them. That is what a robust result looks like: you can attack the poverty line, the reconstruction method, the end date and the units, and the picture does not change shape.

Hold on to that, because the next chart in this chapter does not survive its first caveat at all.

And here is a chart that looks like a pattern

The data for this one

A different data set: a panel of the 50 US states carrying a house price index, its change over the past decade, the unemployment rate, current population, and population growth over the past decade.

	Member	HPI	HPIO	gHPI	gHPI5	nDOWN	UR	POPN	POPO	gPOP	HPI_date	HPIO_date	UR_date	POP_date
0	AL	540.51	286.54	88.6	48.2	5	3.2	5193.088	4854.803	7.0	2026-01-01	2016-01-01	2026-06-01	2025-01-01
1	AK	495.28	310.87	59.3	42.5	7	4.4	737.270	738.430	−0.2	2026-01-01	2016-01-01	2026-06-01	2025-01-01
2	AZ	734.15	331.39	121.5	47.7	3	4.9	7623.818	6832.810	11.6	2026-01-01	2016-01-01	2026-06-01	2025-01-01
3	AR	478.52	254.07	88.3	52.3	1	4.1	3114.791	2979.732	4.5	2026-01-01	2016-01-01	2026-06-01	2025-01-01
4	CA	976.79	543.85	79.6	35.9	5	5.2	39355.309	38904.296	1.2	2026-01-01	2016-01-01	2026-06-01	2025-01-01
5	CO	854.97	445.09	92.1	34.3	5	3.9	6012.561	5454.328	10.2	2026-01-01	2016-01-01	2026-06-01	2025-01-01
6	CT	720.72	384.40	87.5	60.6	5	5.2	3688.496	3588.561	2.8	2026-01-01	2016-01-01	2026-06-01	2025-01-01
7	DE	775.22	427.02	81.5	48.5	2	4.9	1059.952	942.065	12.5	2026-01-01	2016-01-01	2026-06-01	2025-01-01

	HPI	gHPI	UR	POPN	gPOP
count	50.00	50.00	50.00	50.00	50.00
mean	701.43	94.09	3.97	6821.82	6.33
std	214.11	22.04	0.82	7661.85	5.57
min	361.29	42.10	2.00	588.75	−4.20
25%	532.59	80.07	3.32	2020.94	2.32
50%	682.50	92.05	4.10	4612.53	5.50
75%	853.36	112.85	4.57	7906.72	10.72

	HPI	gHPI	UR	POPN	gPOP
max	1312.72	153.20	5.20	39355.31	22.80

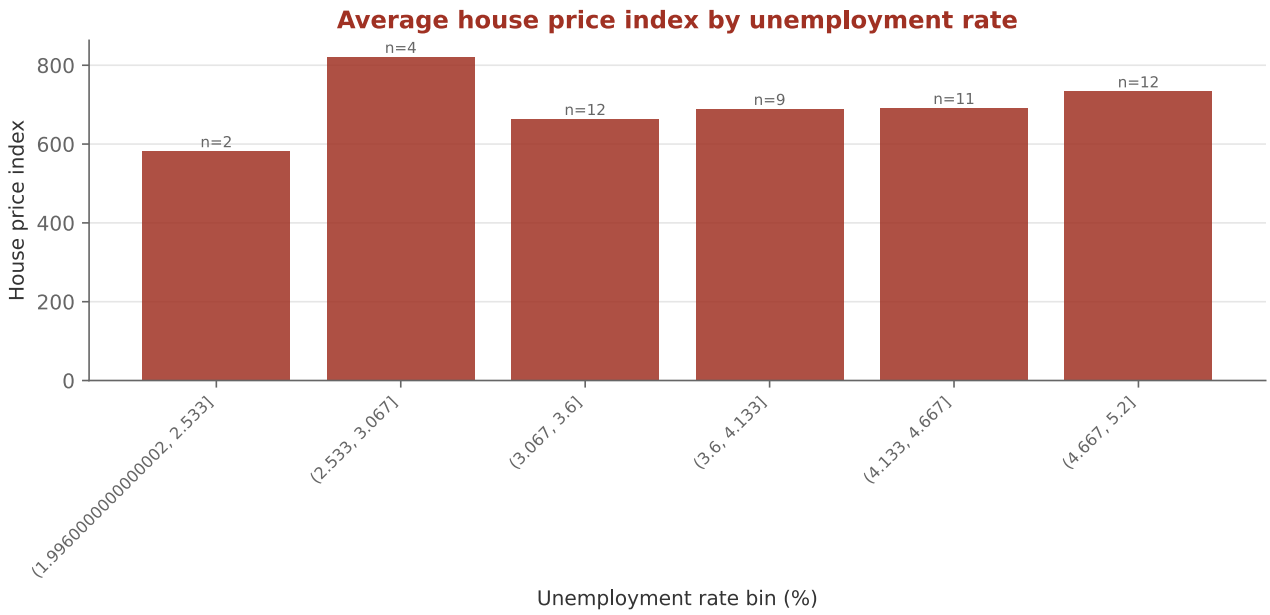
Before reading anything into those numbers, one check carried over from Chapter 1:

```
House price index      as of 2026-01-01  (all states agree)
Unemployment rate      as of 2026-06-01  (all states agree)
Population             as of 2025-01-01  (all states agree)

Both growth rates cover ten years -- but not the same ten years:
  House price growth   2016-01-01 to 2026-01-01
  Population growth    2015-01-01 to 2025-01-01
```

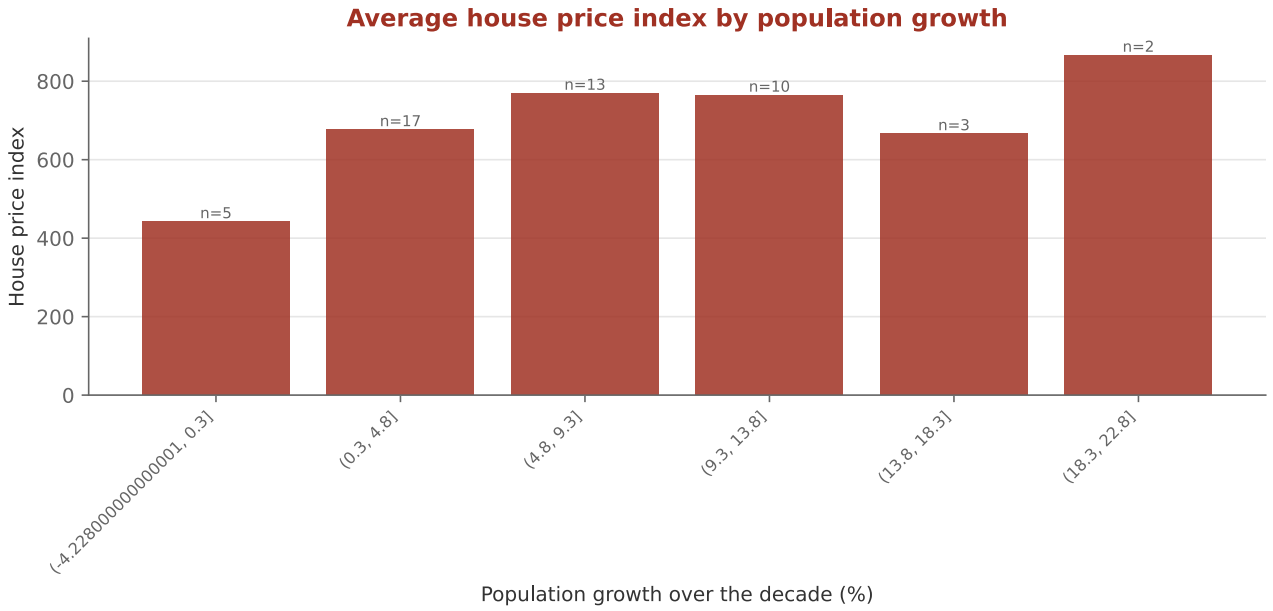
> These variables are measured **as of different dates**, spanning more than a year, and the two ten-year windows are offset from each other by about a year. Every chart in this chapter puts them side by side anyway. That is normal practice, and it is a limitation you should state rather than hide — house prices from one quarter and unemployment from six months later are not a snapshot of a single moment.

Here is average house price index by unemployment rate, in equal-width bins.



There is no pattern here worth reporting. The bars wander. Note the `n=` labels: some bins hold a handful of states, and a bar computed from four observations should not be read as though it were computed from twenty.

Now the same chart against population growth.



This one does look like something: states with faster population growth tend to have higher house price indexes. Before believing it, ask how strong "tend to" is.

```
Correlation of house price INDEX with unemployment rate    r = +0.101
Correlation of house price INDEX with population           r = +0.182
Correlation of house price INDEX with population growth    r = +0.300
```

A correlation of +0.30 is a real relationship and a weak one. That is a fine place to stop — most analyses do. Let us not stop there, because there is something wrong with the question.

Look at which states are at the top

```
Highest house price index
Member     HPI  gPOP
    MA 1312.72   5.2
    NY 1157.14   1.8
    RI 1039.61   5.5
    ME 1035.68   6.5
    WA 1031.17  11.6
    CA  976.79   1.2

Lowest house price index
Member    HPI  gPOP
    WV 361.29  -4.2
    LA 376.99  -1.0
    OK 395.71   5.4
    MS 424.20  -1.2
    IA 462.37   3.7
    AR 478.52   4.5

Fastest-growing states, and where they rank on the index
Member  gPOP    HPI
    ID  22.8 873.60
    UT  18.6 858.12
    FL  16.0 827.88
    TX  15.4 529.31
```

```
    NV  14.4 640.48
    SC  13.8 717.02
```

Massachusetts, New York and Rhode Island lead a chart supposedly about housing, and they are among the slowest-growing states in the country. Texas grew 15.4% and sits in the bottom half. Idaho grew 22.8% — the fastest in the nation — and is nowhere near the top.

Something is wrong. Read the series documentation.

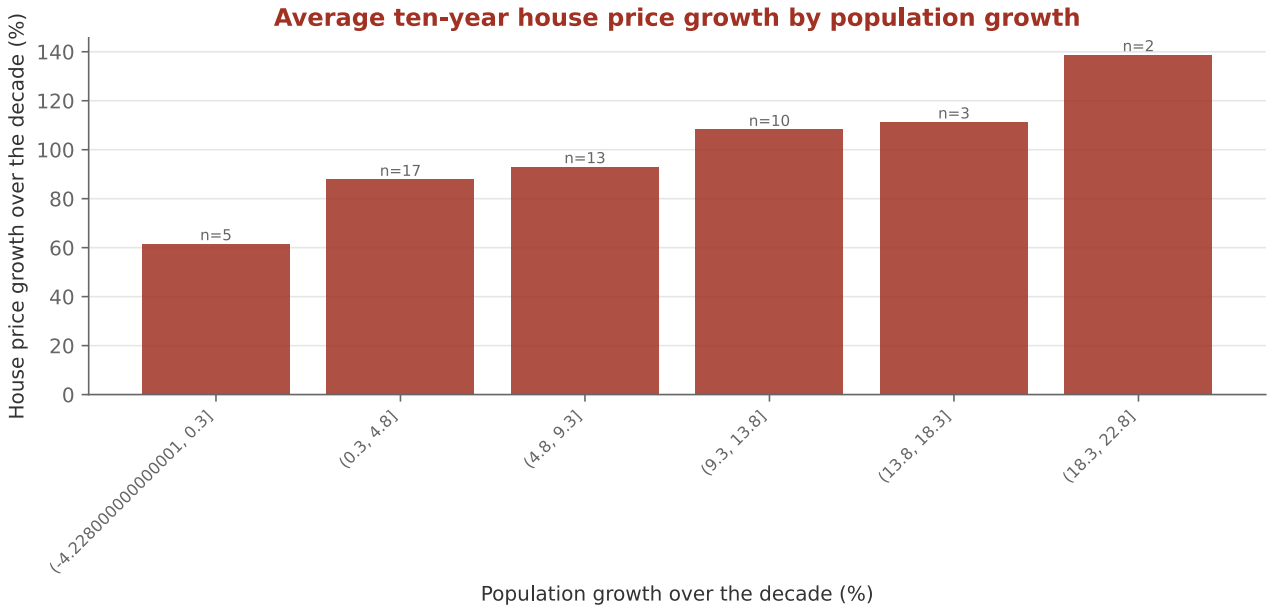
```
Series     : {ST}STHPI, one per state
Name       : All-Transactions House Price Index
Source     : U.S. Federal Housing Finance Agency, via FRED
Units      : Index, 1980 Q1 = 100
Frequency  : Quarterly
```

> **Index, 1980 Q1 = 100.** Every state starts at 100 in 1980 by construction. So a state's index *level* does not measure what a house costs there. It measures **how much prices have risen since 1980** — cumulative appreciation over forty-five years.
>
> Massachusetts is at the top because Massachusetts has appreciated the most since 1980, not because it is expensive today. The chart is not a housing price chart at all. It is a chart of forty-five-year growth, being compared against ten-year growth.

This is the same failure as the consumer confidence series later in this chapter: two correct numbers, describing the same reality, that are not comparable because they were constructed differently. Here it is worse in one way — nothing in the file says so. The column is labelled `HPI` and every value in it is right.

The fix is to compare like with like

`gPOP` is a **ten-year change**. So compare it to the **ten-year change in the index**, which is a percentage and is comparable across states.



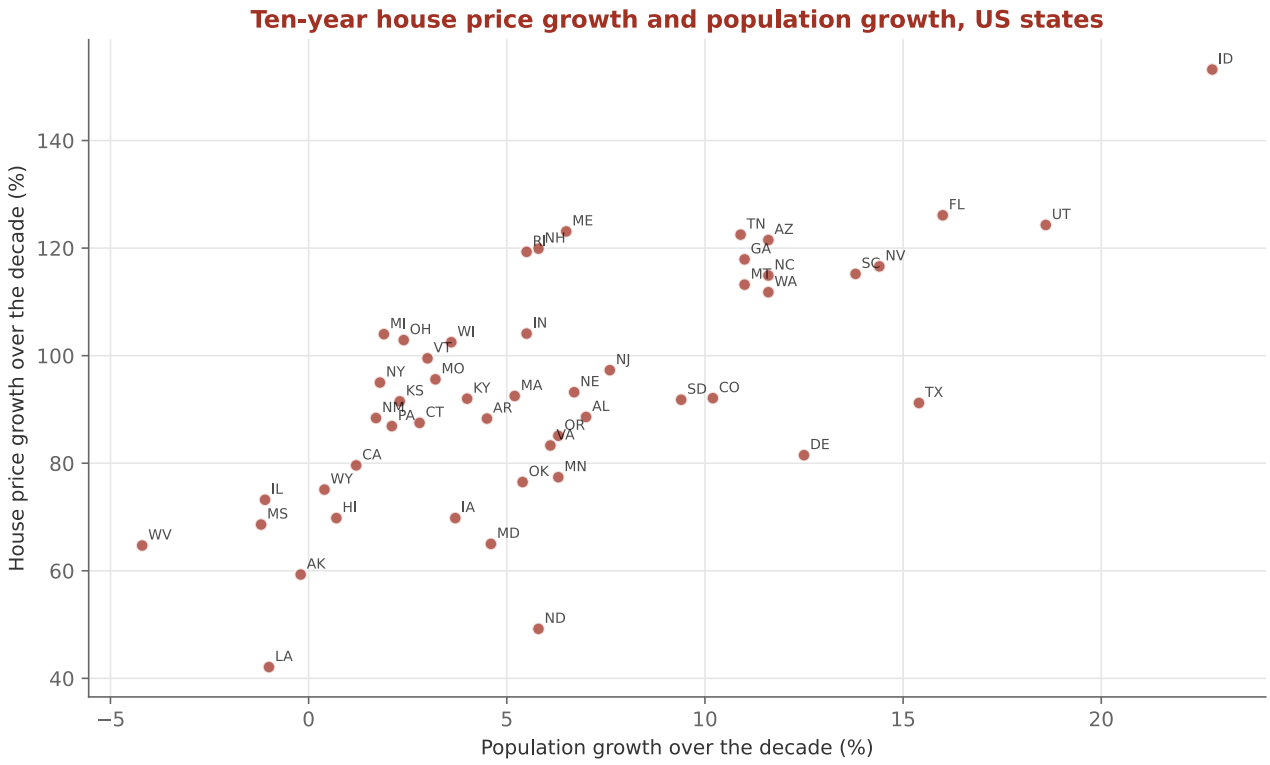
```
house price INDEX LEVEL    vs population growth    r = +0.300
house price CHANGE (10y)   vs population growth    r = +0.705
house price CHANGE (10y)   vs unemployment rate    r = -0.020
```

Same fifty states. Same source. Same download. The correlation went from **+0.30 to +0.71** because the units changed, and nothing else did.

The unemployment rate is still flat against the new variable — r of about zero. That matters: it says the improvement is not an artefact of making everything correlate with everything. One pairing got much stronger and the other did not move.

And now be careful again

A correlation of +0.71 is strong enough that people will start saying causal things about it. It still does not license them.



The scatter shows what the bar chart cannot: real spread, and obvious exceptions. Maine and New Hampshire grew slowly and appreciated enormously. North Dakota grew faster than average and appreciated less than almost anyone. Compare this with the binned bar chart above, where those states disappeared into an average.

Binning averages away the disagreement. Both charts are honest; the scatter is more informative about how much confidence the pattern deserves.

And the strongest claim either one supports is still: *states with faster population growth tended to see faster house price growth over the same decade.* Not that population growth raised prices. People may be moving toward places where prices are rising, prices may be rising where people are moving, or a third factor — climate, taxes, industry, land-use rules — may drive both. **The chart cannot tell you which, and neither can r = 0.71.**

The window is a choice too

One more thing to be suspicious of. Ten years was not handed to us; we picked it. Here is the same comparison over five years.

```
Correlation of population growth with house price growth
  measured over ten years   r = +0.705
  measured over five years  r = +0.159

Spread of house price growth across the fifty states:
  over ten years     42.1% to  153.2%   (standard deviation 22.0)
  over five years    22.7% to   64.9%   (standard deviation 9.1)
```

Over five years the relationship all but disappears — and the second table says why. After 2021 essentially every state's prices rose, and they rose by similar amounts, so there is very little variation left across states for anything to explain.

> A relationship that is strong over one window and absent over another has not been discovered. It has been *found in a particular window*, and the window belongs in the sentence: "over the decade to 2026." Anyone who reports the number without the window is reporting something they cannot support.

Displaying quantitative data

Two kinds of number

Before any of this, a distinction that decides how the rest of the chapter behaves — and which will still be deciding things in Chapter 5 and Chapter 6.

Discrete data are values that have been *counted*. They are restricted to whole numbers. The number of quarters in which house prices fell, the number of employees who left last month, the number of defective units in a batch: you cannot have 3.4 of any of them.

Continuous data are values that have been *measured*. They can take any real number within their range, limited only by the precision of the instrument. Population growth of 6.3%, a hold time of 4.82 minutes, a temperature of 76.4 degrees.

> **The acid test.** Ask: *how many values are possible inside a given interval?*
>
> Between 0 and 3, a count can take four values — 0, 1, 2, 3. A measurement can take infinitely many: 2.0, 2.04, 2.046, and on forever.
>
> A finite answer means discrete. An infinite answer means continuous.

Counted or measured is the question, and it is not always the same as "does the number happen to be a whole number today."

```
Quarters in which a state's house prices fell  discrete    counted; there is no 3.4th quarter
Population growth over the decade (%)          continuous  measured; 6.31 and 6.312 are both possible
Days the US spent in recession                 discrete    counted
Consumer confidence index                      continuous  measured on a scale
Respondents choosing each poll answer          discrete    counted
Unemployment rate (%)                          continuous  a measured proportion
```

The awkward middle. Some variables are technically discrete and are displayed as though they were continuous, and this is normal practice rather than a mistake.

A golf score is counted in strokes, so it is discrete. Annual income is counted in cents. Age is counted in years. But when the range is wide and the steps are tiny relative to it — scores from 279 to 308, incomes from $40,000 to $88,000 — grouping them into classes and drawing them as continuous is both conventional and more readable.

The test that matters in practice is not "is this technically discrete" but **"are there few enough distinct values that each one can be its own class?"** Six values, yes. Three hundred, no.

A frequency distribution with one class per value

When the distinct values are few, each value gets its own class and nothing is grouped at all. Here is a genuine count, from the same fifty states.

For each state, we took the last ten years of the quarterly house price index — forty quarterly changes — and counted how many of them were **negative**. How many times in a decade did that state's housing market go backwards?

	Category	Frequency	PCT	CUM_FREQ	CUM_PCT
0	0	2	4.0	2	4.0
1	1	13	26.0	15	30.0
2	2	9	18.0	24	48.0
3	3	6	12.0	30	60.0
4	4	5	10.0	35	70.0
5	5	9	18.0	44	88.0
6	6	0	0.0	44	88.0
7	7	3	6.0	47	94.0
8	8	1	2.0	48	96.0
9	9	2	4.0	50	100.0

No bins were chosen, no width was decided, no boundary had to be argued over. The values *are* the classes. This is the simplest frequency distribution there is, and it is the one the textbook builds first.

Read it straight off. **Thirty of the fifty states — 60% — had three or fewer falling quarters in a decade.**

```
n = 50 states, 0 to 9 falling quarters
mean 3.22   median 3

Never fell:  KY, SC
Fell most:   HI, ND (9 of 40 quarters)

States with exactly 6 falling quarters: none
```

The class that isn't there

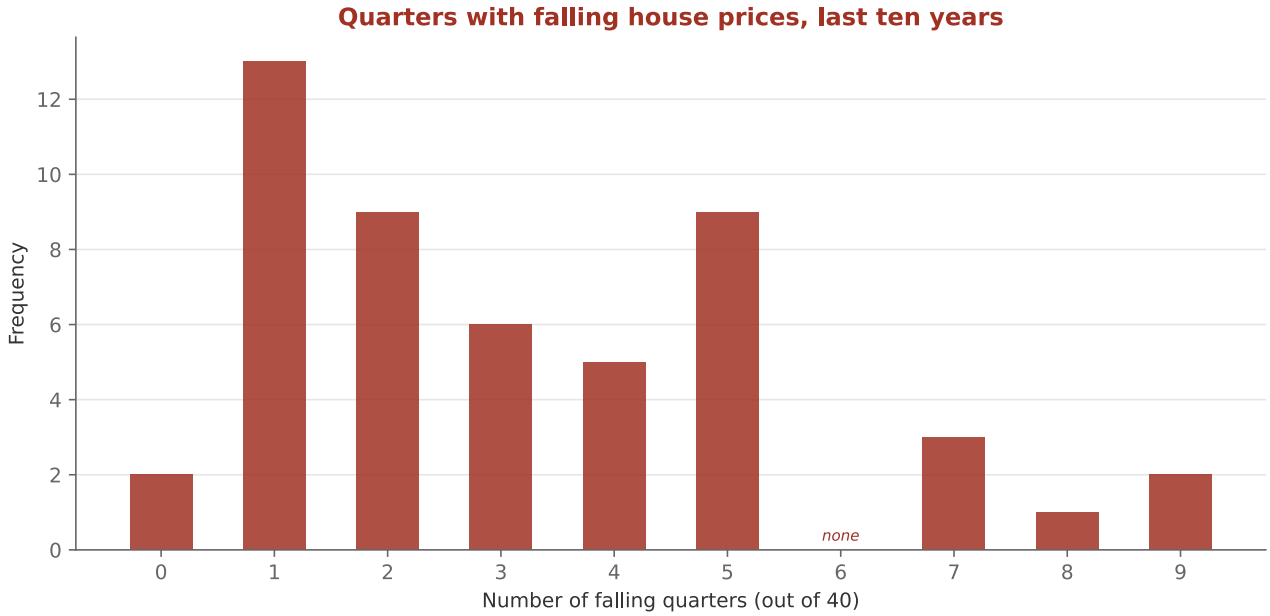
Look again at the row for **6**. Its frequency is zero.

That row is easy to delete and tempting to delete, and deleting it would be wrong. A class with no observations in it is a *finding* — it says something about the shape of the distribution. Leave it in, and the reader can see that

the distribution has a gap in it. Take it out, and the 5 and the 7 sit next to each other as though nothing were missing.

> This is the same instinct as Tukey's rule for stem and leaf displays, later in this chapter: **don't cut your leafless stems.** An empty class and an empty stem are the same thing, and both should stay.
>
> It is also the fourth rule of building grouped classes — *avoid empty classes* — seen from the other side. That rule is about not choosing a class width so narrow that you manufacture empty classes. This one is about not hiding an empty class you did not manufacture.

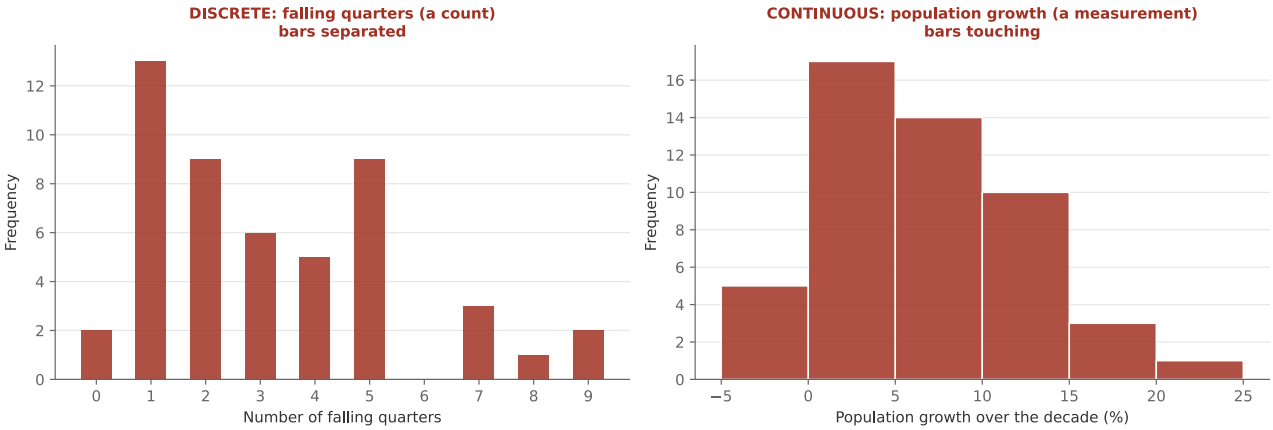
Drawing it: the gap between the bars



Notice the **gaps between the bars**. That is not a styling choice. It is a convention with a meaning:

Data	Bars	Because
Discrete	separated	nothing exists between 2 and 3, so nothing is drawn there
Continuous	touching	every value between 2 and 3 exists and belongs to some class

Here are both, side by side, from this same data set.



A reader who knows the convention learns what kind of variable they are looking at before reading a single label. A reader who does not still gets the counts right. **You lose nothing by following it and you mislead people by breaking it** — Excel, left alone, will draw a continuous variable with gaps and a discrete one without, and it is on you to fix it.

> **Why this outlives Chapter 2.** Counted-or-measured is not a charting detail. It is the fork the rest of the course runs on.
>
> **Chapter 5** builds probability distributions for *discrete* variables, where you can ask "what is the probability of exactly 3?" and get a real answer. **Chapter 6** builds them for *continuous* variables, where the probability of exactly any single value is zero and every question has to be asked about an *interval* instead.
>
> The same question — "what is the chance of getting 3?" — is sensible in one chapter and meaningless in the other. Which chapter you are in is decided by the distinction on this page.

When the values have to be grouped

Fifty states produced counts spanning ten whole numbers, nine of which actually occurred — few enough that one class per value worked. Now take a *measured* variable from the same file.

```
Falling quarters (counted)          9 distinct values among 50 states
Population growth % (measured)     44 distinct values among 50 states
```

Population growth has almost as many distinct values as it has states. One class per value would give a table of forty-odd rows, most of them holding a single state, which is a list of the raw data with extra steps.

So the values get **grouped**: several of them share a class.

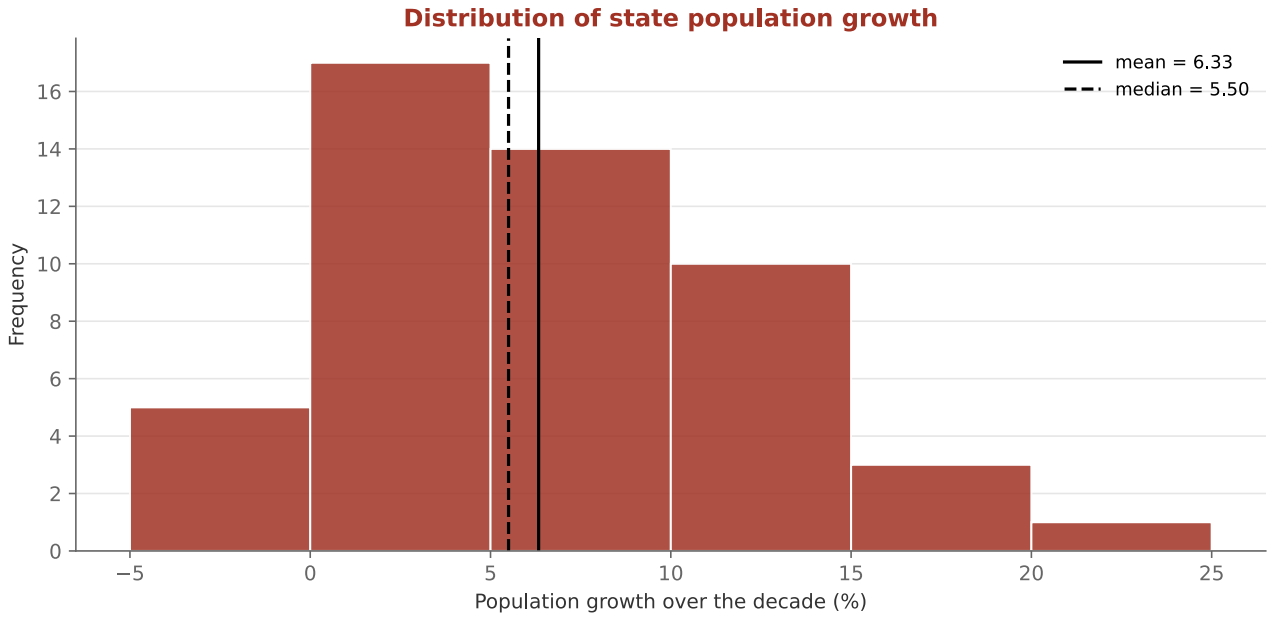
A **frequency distribution** counts how many values fall in each range, or *bin*. Here is population growth across the 50 states.

	Category	Frequency	PCT	CUM_FREQ	CUM_PCT
0	(−5.001, 0.0]	5	10.0	5	10.0
1	(0.0, 5.0]	17	34.0	22	44.0
2	(5.0, 10.0]	14	28.0	36	72.0
3	(10.0, 15.0]	10	20.0	46	92.0
4	(15.0, 20.0]	3	6.0	49	98.0
5	(20.0, 25.0]	1	2.0	50	100.0

The **PCT** column is the *relative frequency*: the count in each class divided by the total. There are 50 states, so a class holding 17 of them is $17/50 = 34\%$.

The **CUM_PCT** column is the *cumulative relative frequency*, totalling as you move down. Its interpretation is direct: **44% of states grew by 5% or less over the decade, and 72% grew by 10% or less.** Only one state exceeded 20%.

A **histogram** draws the same information.




```
mean     6.33
median   5.50
The mean sits above the median, by 0.83 points.

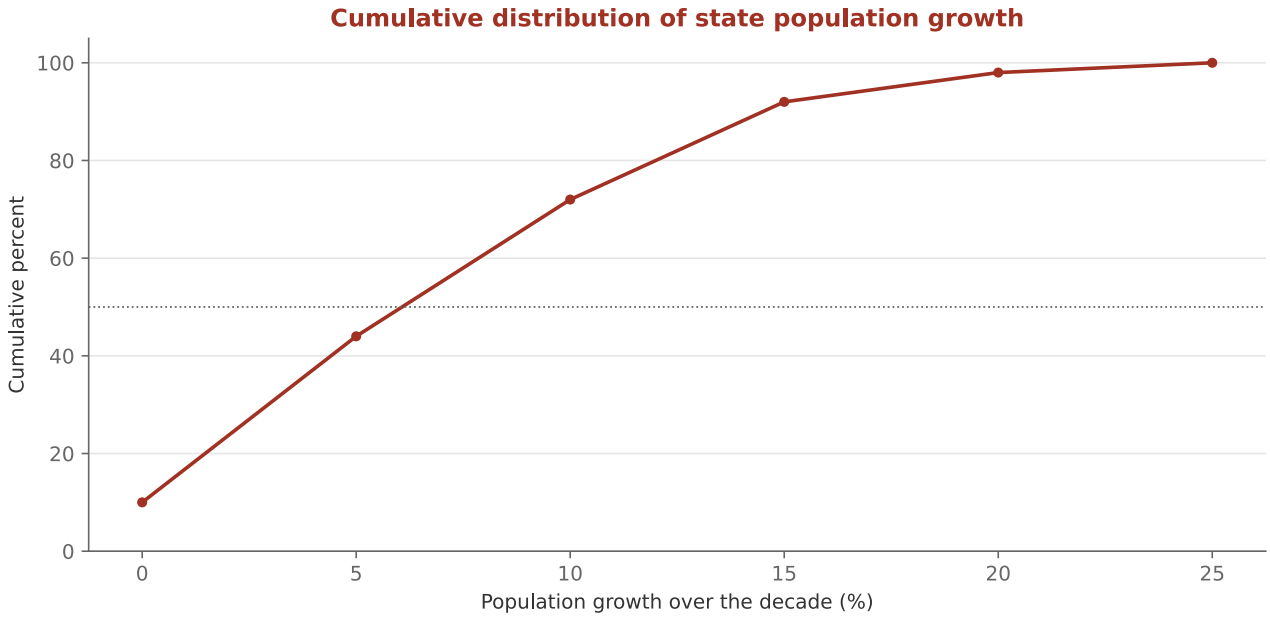
Fastest growing: ID at 22.8%
Slowest growing: WV at -4.2%
```

This is an **asymmetrical** distribution. A *symmetrical* one would be a mirror image on either side of the middle. Here most of the mass sits at the left and a thin tail stretches right — a few fast-growing states pull the mean above the median. That gap between mean and median *is* the skew, expressed as a number.

> When a distribution is skewed, the mean and the median answer different questions. The median describes the typical state. The mean is pulled by the extremes. Reporting only one, without saying which, is how a summary becomes misleading without containing a false number.

The ogive

The **ogive** plots the cumulative relative frequency. A point on the line reads: this percent of observations fall at or below that value.



The dotted line marks 50%. Where the curve crosses it is the median. A steep section means many observations packed into a narrow range; a flat section means few.

How many classes?

For grouped data we have to choose the number of classes. A common rule-of-thumb is to find k such that

$$2^k \geq n,$$

where n is the number of observations and k the number of classes. With 50 states, $2^5 = 32$ and $2^6 = 64$, so $k = 6$.

The class width then follows from the range divided by the number of classes. This is a *rule of thumb*, not a law — and the choice has consequences, as the next section shows.

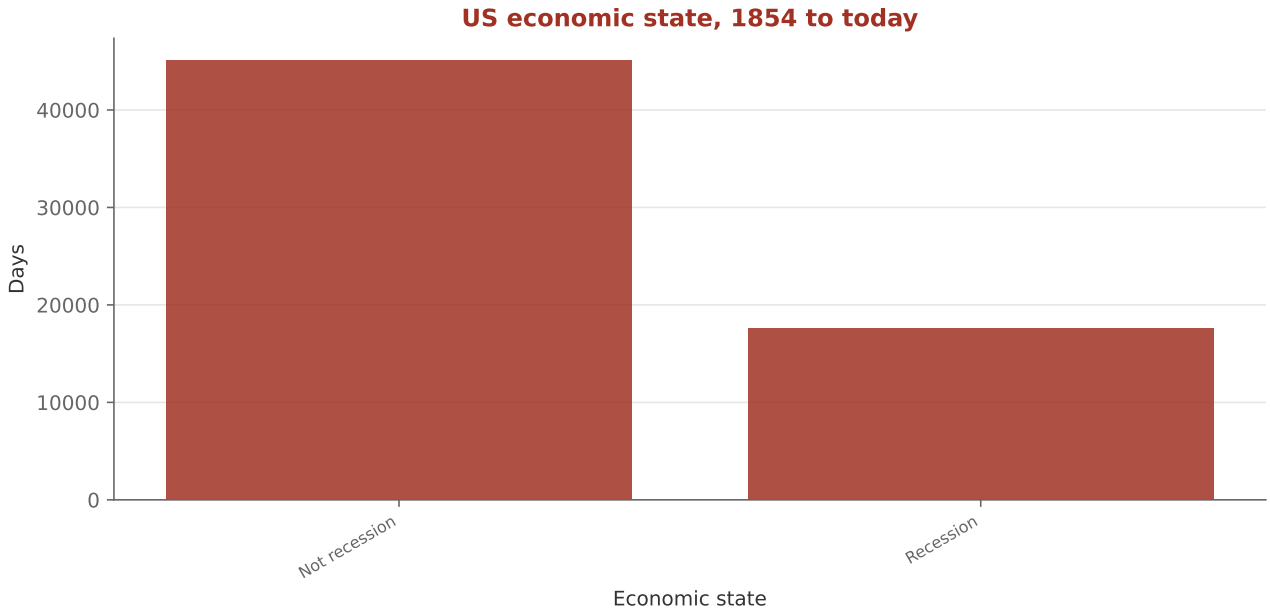
Rules when constructing grouped classes

1. **Equal-size classes.** Unequal widths make patterns misleading.
2. **Mutually exclusive classes.** Boundaries must not overlap.
3. **Include all data values.** Fifty observations, fifty observations placed.
4. **Avoid empty classes.** Too-narrow widths produce classes with no data.
5. **Avoid open-ended classes** where possible — though sometimes an age class of "19+" beats fifty empty classes between 19 and 70.

Displaying qualitative data

From December 1854 the NBER recession indicator marks every day as recession or not. That is *qualitative* data: each day carries a label.

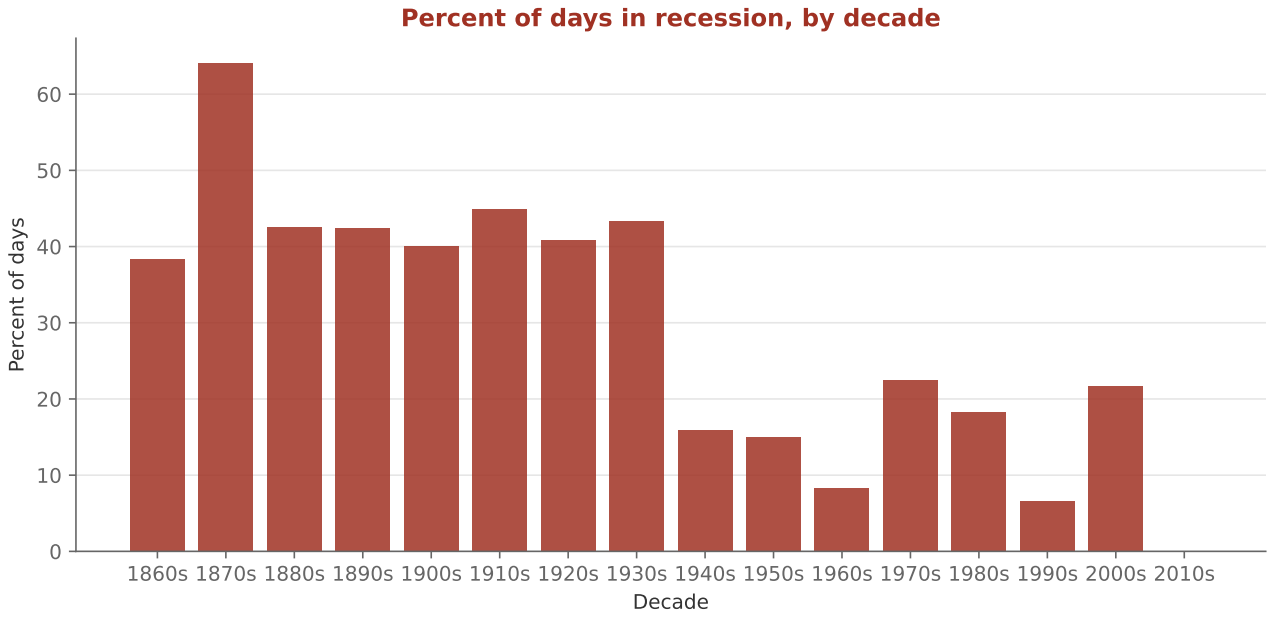
```
62,716 days, 1854-12-01 to 2026-08-16
Days in recession: 17,615 (28.1%)
```



A single bar chart of the whole period hides the more interesting fact.

```
1854 - 1939     43.8% of days in recession
1980 - present  10.4% of days in recession

Ratio: 4.2 times more likely before 1940.
```



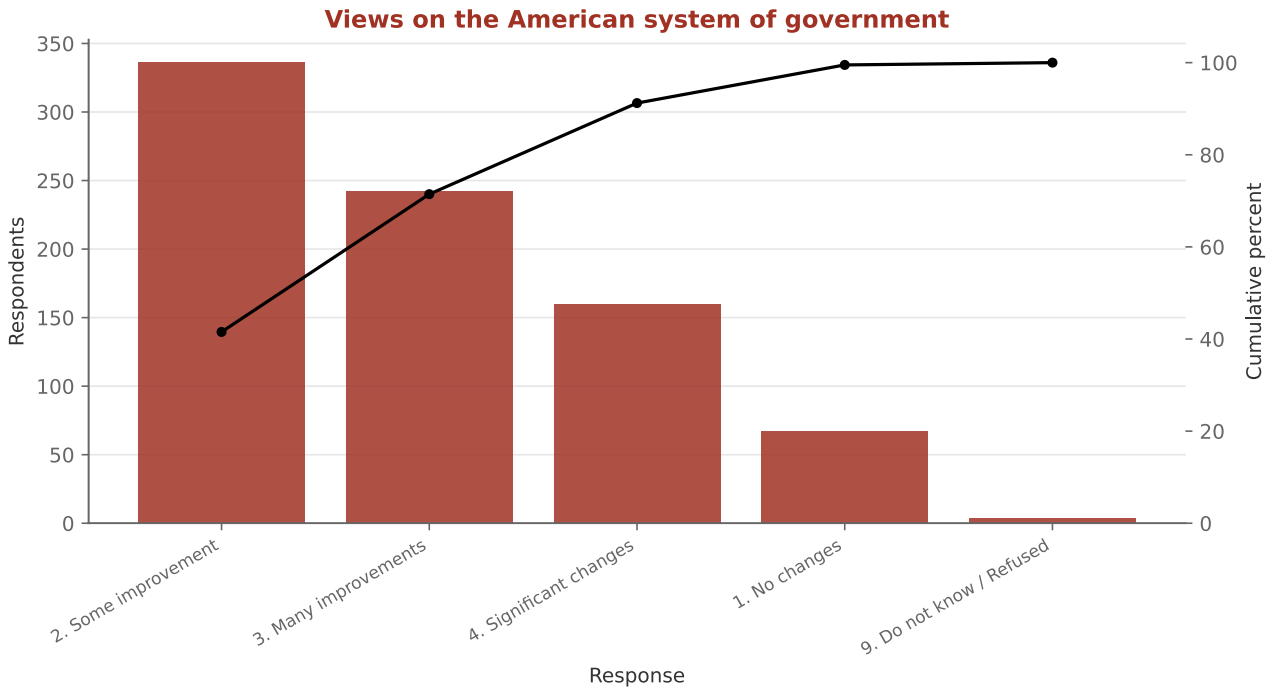
The 1870s stand out at 64% — the Long Depression. The 2010s recorded zero recession days. Reading a chart well means noticing both the shape and the outliers, then asking what produced them.

Pareto charts

A **Pareto chart** is a bar chart sorted from most to least frequent, with a cumulative percentage line. Named for Vilfredo Pareto, it answers “how few categories account for how much?”

We use the *Monmouth University National Poll #240*, January 2021. Question 16 asked which statement came closest to the respondent's view:

1. Our system of government is basically sound and essentially needs no changes
2. Our system is basically sound, but needs some improvement
3. Our system is not too sound and needs many improvements
4. Our system is not sound at all and needs significant changes



Read the black line, not just the bars. The two most common answers account for over three-quarters of respondents. A summary that reported only the modal response would miss that the second category is nearly as large.

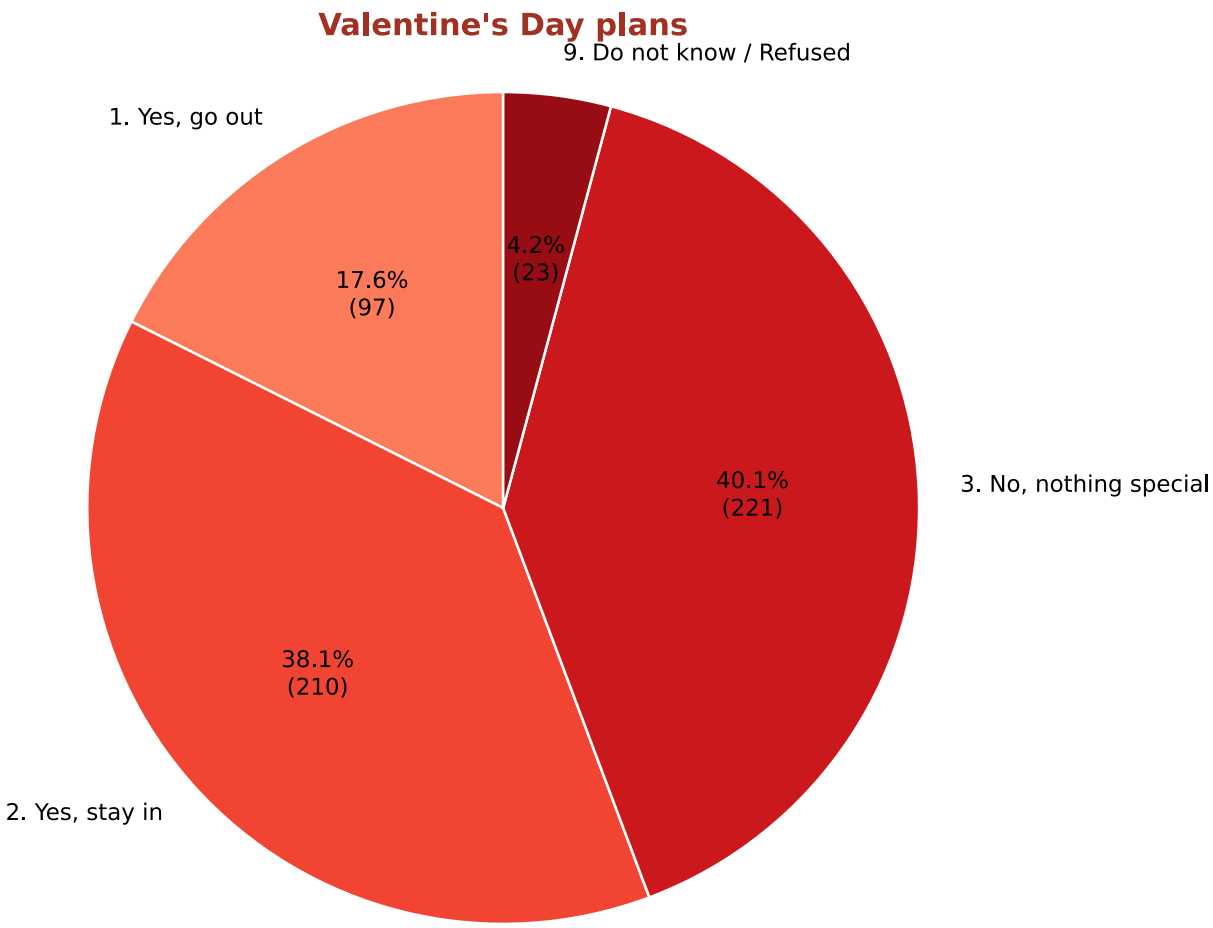
Pie charts, and a trap

Pie charts have been the subject of a smear campaign. The criticism has merit: beyond three or four slices, people cannot compare angles reliably.

Question 38 asked: *Looking ahead to Valentine's Day next month, will you and your partner do anything special to mark the day?*

```
NOTE: 258 observation(s) had no value and are excluded from this table. The percentages below are of the 551 that did.
```

	Category	Frequency	PCT	CUM_FREQ	CUM_PCT
0	1. Yes, go out	97	17.6	97	17.6
1	2. Yes, stay in	210	38.1	307	55.7
2	3. No, nothing special	221	40.1	528	95.8
3	9. Do not know / Refused	23	4.2	551	100.0



Look at what that chart is not telling you. Question 16 was answered by all 809 respondents. Question 38 was answered by 551. The other 258 were never asked — the question presupposes a partner.

```
Q16 (system of government)       answered  809   not answered    0
Q38 (Valentine's plans)          answered  551   not answered  258
```

Every number in that pie chart is correct. Every percentage is correctly computed. And a reader who sees it captioned “Valentine’s Day Plans” will conclude something about *Americans*, when it describes only the 68% who were asked. **The chart does not lie. The caption does.**

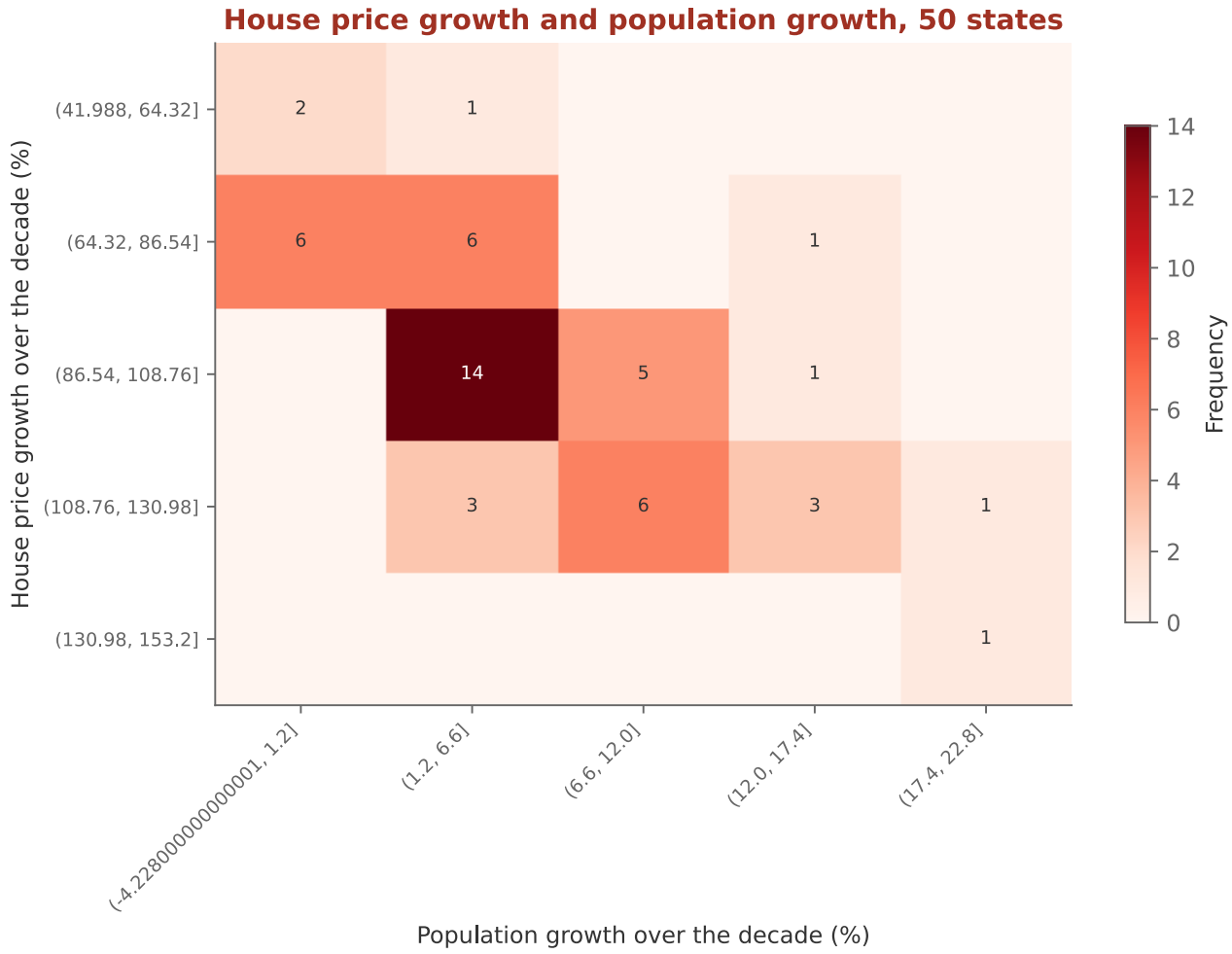
The honest version says what the base is: *“Among the 551 respondents with a partner…”*

Contingency tables

A **contingency table** shows frequencies for two variables at once.

Population growth (%)	(−4.228000000000001, 1.2]	(1.2, 6.6]	(6.6, 12.0]	(12.0, 17.4]	(17.4, 22.8]
House price growth (%)					
(41.988, 64.32]	2	1	0	0	0
(64.32, 86.54]	6	6	0	1	0
(86.54, 108.76]	0	14	5	1	0

Population growth (%)	(−4.228000000000001, 1.2]	(1.2, 6.6]	(6.6, 12.0]	(12.0, 17.4]	(17.4, 22.8]
House price growth (%)					
(108.76, 130.98]	0	3	6	3	1
(130.98, 153.2]	0	0	0	0	1



Colour intensity shows where states concentrate. With 50 observations spread over 25 cells, most cells hold zero, one, or two states — which is itself the finding. **A heatmap of a small sample looks authoritative and carries very little information.** Check the counts before reading a pattern into the colours.

Stem and leaf display

John Tukey originated the **stem and leaf display** in the 1970s, when it could be built by hand. Its virtue is that every individual observation stays visible, unlike a histogram.

One rule carries over from the empty class earlier in this chapter: **do not cut your leafless stems.** If no observation falls on a stem, the stem still appears, empty. Removing it closes a gap that is really there and misstates the shape of the distribution.

```
consumer_confidence.csv
   rows              798
   first observation 1960-01-01
```

```
   last observation  2026-06-01
   any missing       0

consumer_confidence_discontinued.csv
   rows              769
   first observation 1960-01-01
   last observation  2024-01-01
   any missing       0
```

Two files. Same concept — US consumer confidence, monthly, from 1960. Both downloaded without error. Both parsed cleanly. Neither has a missing value. Every number in both is correct.

One of them stopped being published.

```
consumer_confidence.csv                 CURRENT
consumer_confidence_discontinued.csv    STALE by 34 months
```

> Nothing in the file contents distinguishes a live series from a dead one. Row counts look similar, values look plausible, no errors are raised. **The last observation date is the only tell**, and checking it is the verification almost nobody performs.
>
> A retired series produces a perfectly reasonable-looking analysis of a world that no longer exists.

The same thing, measured on two different scales

The two series do not merely resemble each other. Over the months they both cover, they move together almost perfectly.

```
Overlapping months: 769
Correlation over the overlap: 0.992

                current     retired
minimum           48.20       96.20
mean              91.43      100.00
maximum          120.51      102.85

The same four months, read from each series:
      date  cci_new  cci_old
2023-10-01 68.64521 97.51254
2023-11-01 65.95535 97.56221
2023-12-01 74.99328 98.10970
2024-01-01 84.99955 98.91297
```

A correlation of 0.992 says these two series carry the same information. The summary table says they are built differently. The retired series is a *standardised index*, normalised to a long-run mean of exactly 100, never straying more than about three points from it. The current series is a **percentage balance** — roughly, the share of optimistic responses minus the share of pessimistic ones — and it ranges from the high forties to above 120.

These are not the same measurement rescaled. They are two different ways of turning the same survey into a number.

> In October 2023 one of these series reads about 69 and the other about 98. Same month, same concept, same underlying survey.

A student who saw 98 in an old report and 69 in a new one would conclude that consumer confidence had collapsed. Nothing collapsed. **The normalisation changed.**

This is why *units* belong in the provenance record alongside the series name and the date range. “Consumer confidence was 69” is not a fact until you say which measure, constructed how. Two correct numbers describing the same reality can differ by thirty points, and neither is wrong.

```
Series     : USACSCICP02STSAM
Name       : Consumer Opinion Surveys: Composite Consumer Confidence,
             United States
Source     : OECD, via Federal Reserve Economic Data (FRED)
Units      : Percentage balance, seasonally adjusted
Frequency  : Monthly
Date range : 1960-01-01 to 2026-06-01
Rows       : 798
```

An observation worth investigating

```
Lowest reading in the series: 48.20 in May 2026
That is the lowest of 798 monthly observations going back to 1960.

Ten lowest readings on record:
   month      cci
May 2026 48.20228
Jun 2026 53.25921
Apr 2026 53.58200
Jun 2022 53.79718
Nov 2025 54.87313
Jul 2022 55.41110
May 1980 55.62629
Apr 2025 56.16426
May 2025 56.16426
Apr 1980 56.70223
```

Consumer confidence is at its lowest recorded level in more than sixty years of this survey — below 2008, below 2020. Look at where the other low readings cluster and what was happening then.

A question to think about. Consumer confidence measures how households *feel* about the economy. Other indicators — employment, asset prices, output — measure what is *happening* to it. These usually move together. When they diverge sharply, one of three things is true: the sentiment measure is picking up something the hard data has not registered yet, the hard data is capturing something sentiment ignores, or the two are responding to different events entirely.

Which chart would you build to tell them apart? What would you need on the same axis, and over what period?

The discontinued series is `CSCICP03USM665S`; the current one is `USACSCICP02STSAM`. If you had built a forecast on the first one in early 2024 it would have looked fine, right up until someone asked what happened last quarter.

We use the current series from here on.

```
Consumer confidence, June 2024 to June 2026

   48 | 2
```

```
  49 |
  50 |
  51 |
  52 |
  53 | 3 6
  54 | 9
  55 |
  56 | 2 2 9
  57 | 3 7
  58 |
  59 | 3
  60 | 7 9
  61 | 3
  62 | 6
  63 |
  64 |
  65 | 3
  66 | 4
  67 |
  68 |
  69 | 6
  70 |
  71 | 4
  72 |
  73 | 1 4
  74 |
  75 | 4 9
  76 |
  77 | 1 3
  78 |
  79 | 6

25 observations. Stem = whole number, leaf = first decimal place.
```

Read down the stems to see where readings cluster. Unlike a histogram, every individual observation is still there — you can recover the actual values, which is why Tukey liked it and why it survives in exploratory work.

```
Lowest reading:  48.20 in May 2026
Highest reading: 120.51 in January 2000
Most recent:     53.26 in June 2026
```

Scatter plots

A **scatter plot** displays two variables with no summarising at all. Every observation is shown. Most analysts draw one first when meeting a new data set.

Which variable goes on which axis

Before drawing one you have to decide which variable goes where, and the convention carries a claim.

The **independent variable** goes on the horizontal axis. It is the one you are treating as given. The **dependent variable** goes on the vertical axis. It is the one you believe responds:

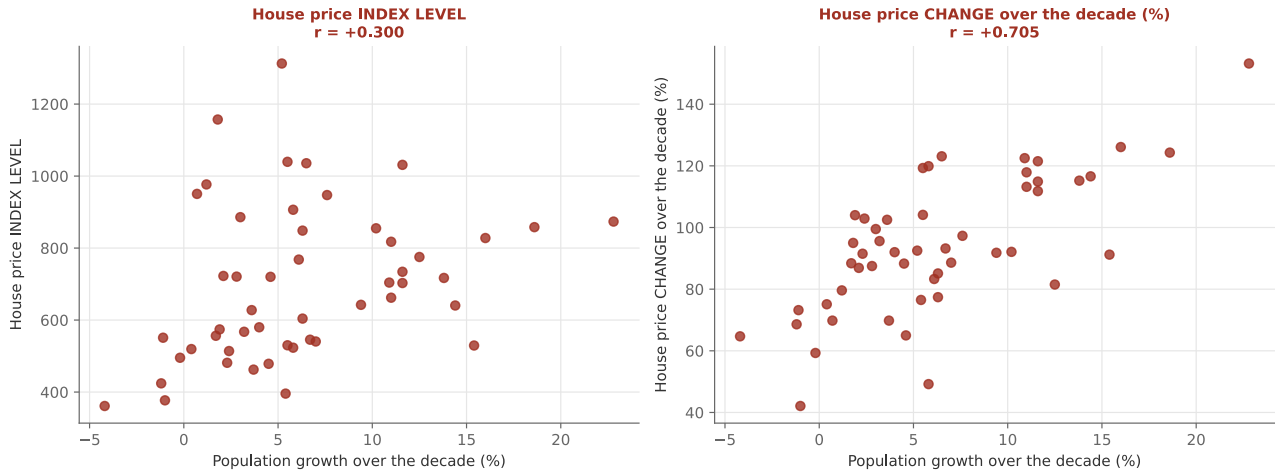
$$\text{independent } (x) \rightarrow \text{dependent } (y)$$

The arrow only points one way. The textbook's example is television size and price: bigger televisions cost more, so size is x and price is y — and cutting the price of a television does not shrink it.

Notice what you have just done. Putting population growth on the horizontal axis and house price growth on the vertical is a *decision about which one you think responds to which*, and it is made before any statistics happen. Nothing in the data forces it. Someone who believes people move toward places where prices are already rising would draw the axes the other way round and get an identical-looking cloud with an identical correlation.

The axis convention is where regression begins — Chapter 10 will ask you to defend the choice rather than assume it. For now: make the choice deliberately, and say out loud that you made it.

The same states, two variables, one substitution

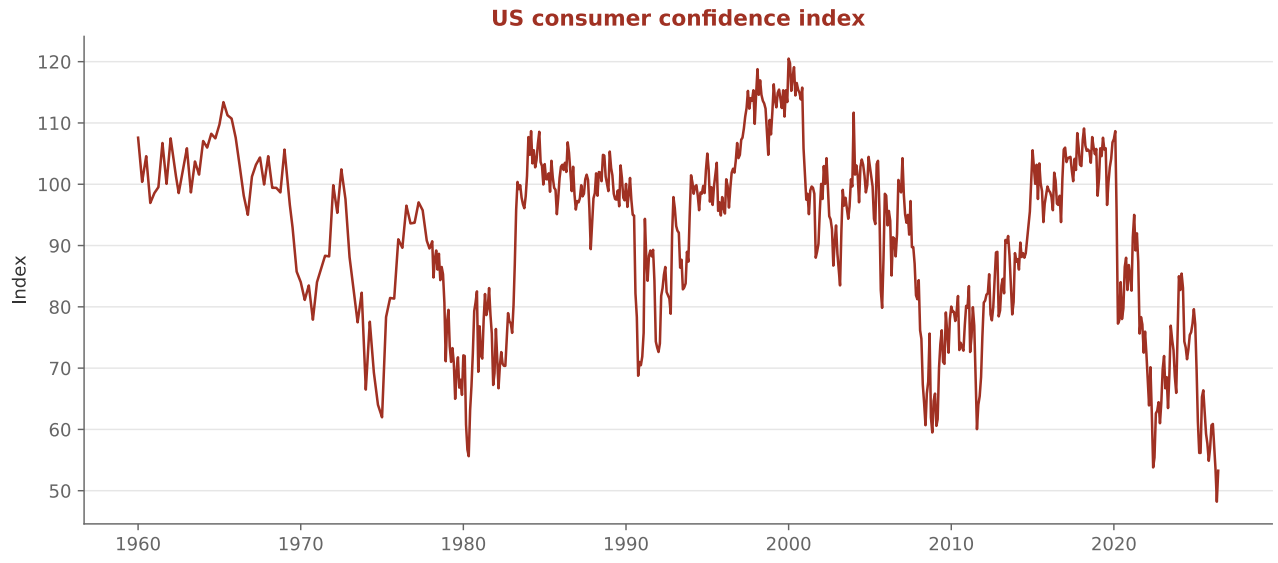


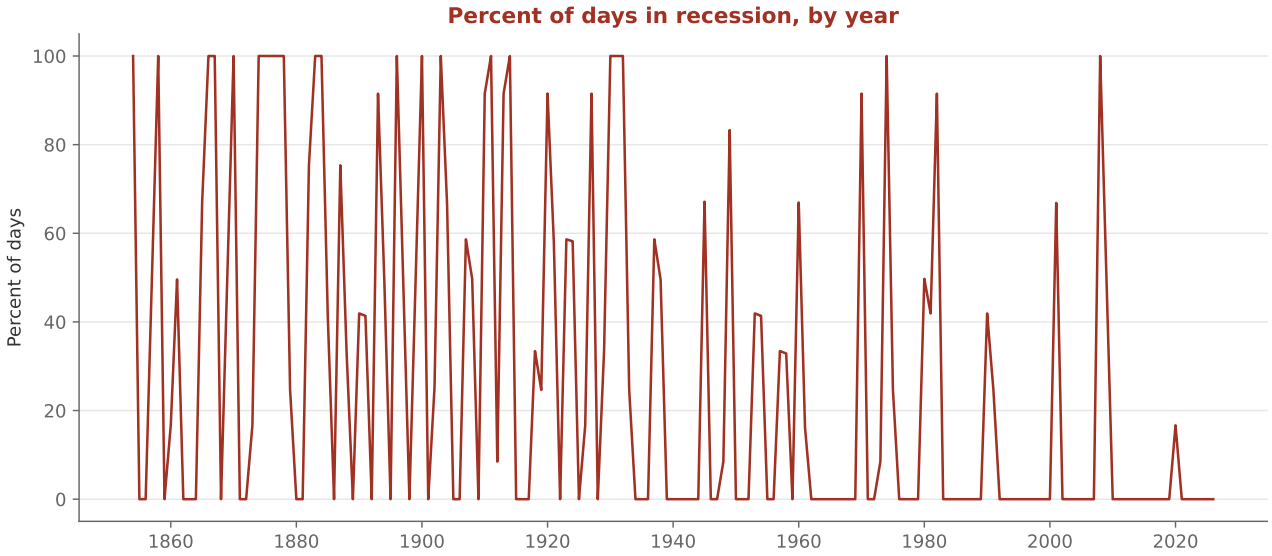
The left panel is a cloud. The right panel is a relationship. The horizontal axis is identical in both; only the vertical variable changed, and it changed from an index level to a percentage change of that same index.

A scatter plot cannot rescue a badly chosen variable — but it shows you faster than anything else that you have chosen one. This is why analysts draw scatter plots first.

Time series

A scatter plot where one variable is time shows a *time path*. Connecting the points produces a **line chart**, which is normally the right way to show a time series.

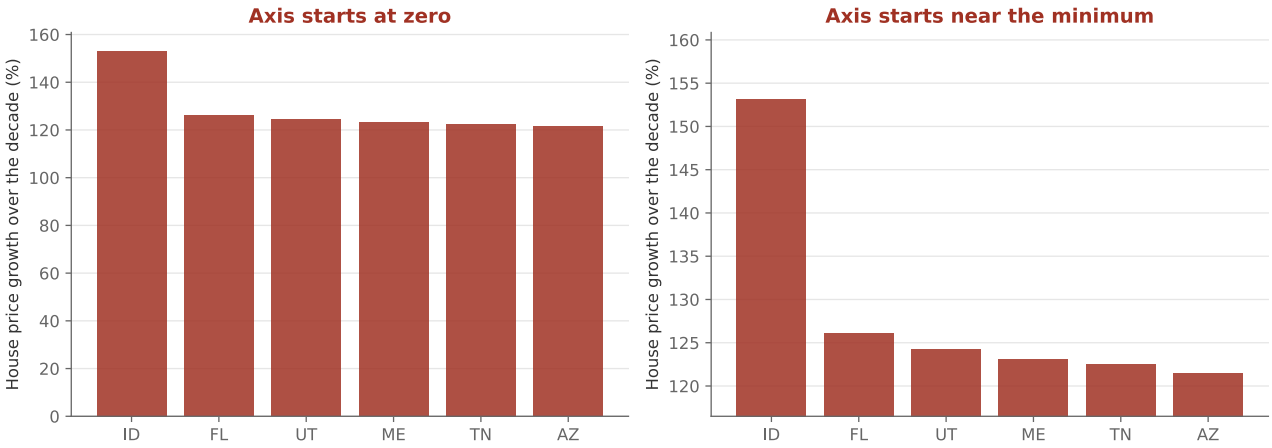




Charts that mislead

Everything below is technically correct and still misleading. These are the failures to watch for — in other people's charts, and in your own.

Truncating the axis




```
Highest: 153.2%   Lowest of these six: 121.5%
Actual gap between them: 31.7 percentage points
```

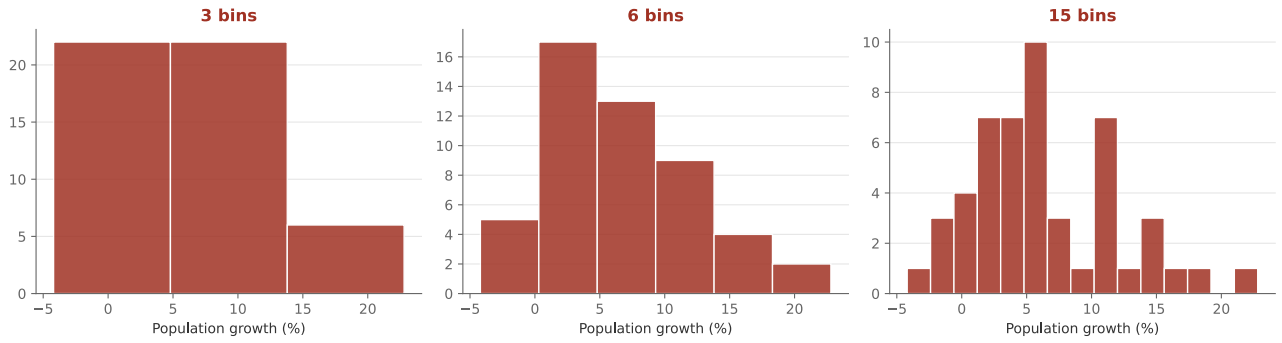
Same six states, same six numbers. The right-hand chart makes the gap look enormous. For bar charts, where length encodes the value, the axis should start at zero.

Letting bin width choose the story

```
--- 3 bins ---
                Category  Frequency  PCT
(-4.228000000000001, 4.8]         22 44.0
             (4.8, 13.8]         23 46.0
            (13.8, 22.8]          5 10.0
```

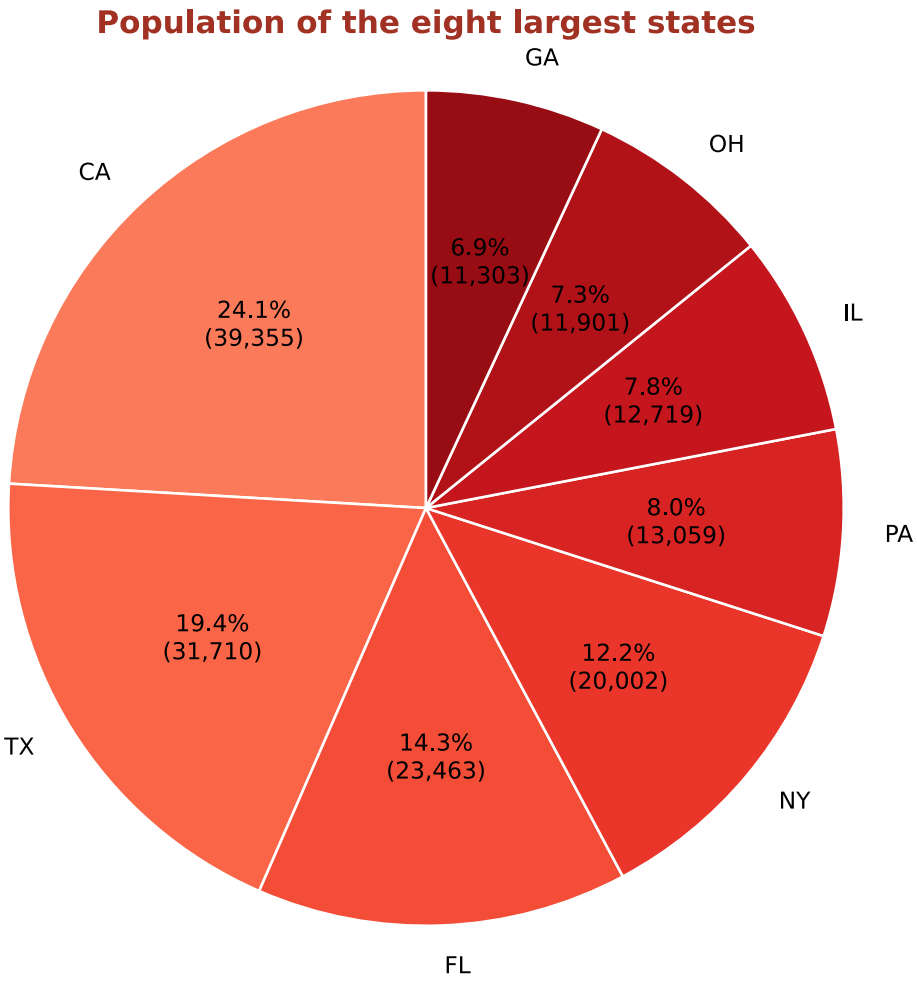
```
--- 6 bins ---
                 Category  Frequency  PCT
(-4.228000000000001, 0.3]          5 10.0
               (0.3, 4.8]         17 34.0
               (4.8, 9.3]         13 26.0
              (9.3, 13.8]         10 20.0
             (13.8, 18.3]          3  6.0
             (18.3, 22.8]          2  4.0

--- 15 bins ---
                 Category  Frequency  PCT
(-4.228000000000001, -2.4]          1  2.0
              (-2.4, -0.6]          3  6.0
               (-0.6, 1.2]          4  8.0
                (1.2, 3.0]          8 16.0
                (3.0, 4.8]          6 12.0
                (4.8, 6.6]         10 20.0
                (6.6, 8.4]          3  6.0
                (8.4, 10.2]         2  4.0
              (10.2, 12.0]          6 12.0
              (12.0, 13.8]          2  4.0
              (13.8, 15.6]          2  4.0
              (15.6, 17.4]          1  2.0
              (17.4, 19.2]          1  2.0
              (19.2, 21.0]          0  0.0
              (21.0, 22.8]          1  2.0
```



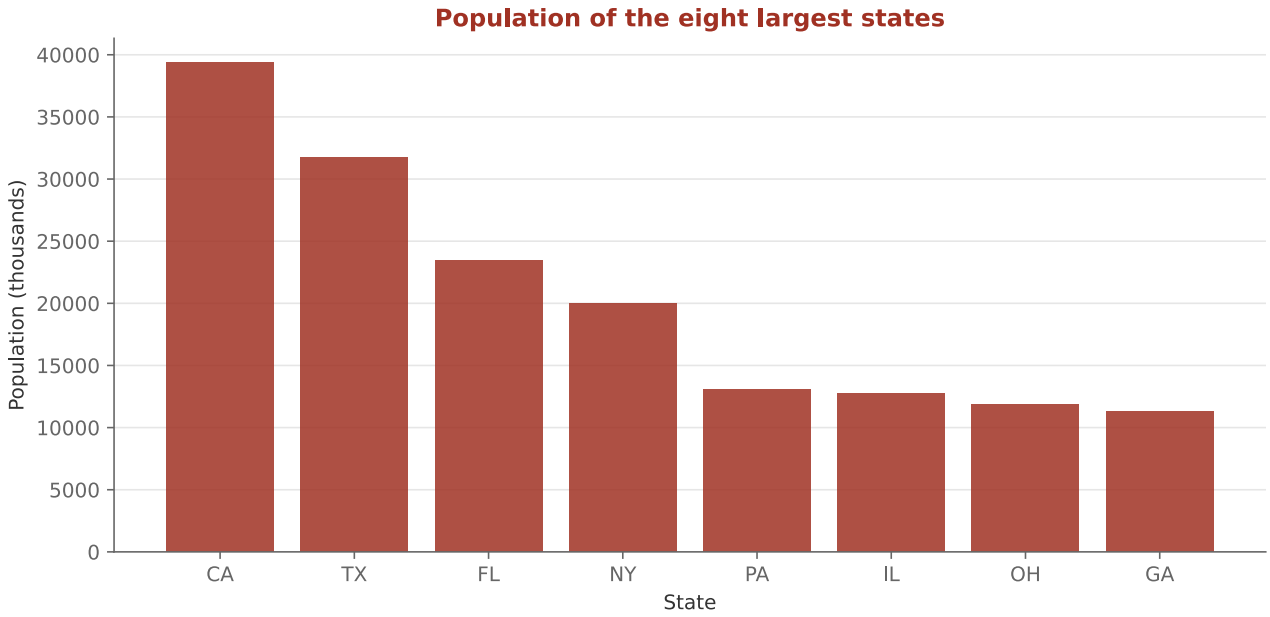
Three bins say the distribution is smooth. Fifteen bins say it is lumpy and possibly bimodal. Same fifty states. **Nobody looking at one histogram can tell you which bin width you tried first.**

The wrong chart for the data




```
NOTE: 8 slices. Readers cannot reliably compare more than three or four angles -- consider a bar chart.
```

Eight slices. Try ranking the third, fourth, and fifth largest by eye, then check against the bar chart.



What to check in any chart

1. **Does the axis start at zero?** Required for bars; often wrong for lines.
2. **What is the base?** Percent *of what*, and who was excluded.
3. **How many observations per bar or cell?** A tall bar over four observations is noise.
4. **Would a different bin width tell a different story?**
5. **How old is the data?** Check the last observation date.
6. **Does the caption claim more than the chart shows?** Correlation drawn as a chart is still correlation.

Laboratory 2

There are two ways to get these files to Claude. **Start with Path 1.** It works on every Claude account, including the free tier, and needs nothing turned on.

Path 1 — upload the files

Download the Laboratory 2 pack — one file. Unzip it and you will have a folder holding 7 files. Drag all 7 into a new Claude conversation, wait for each to show as attached, then paste this:

PROMPT

> I have uploaded 7 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.
>
> Then run `02_displaying_data.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

If web access is working on your account, Claude can retrieve the files itself:

PROMPT

Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 2 and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.

Then run `02_displaying_data.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach GitHub, **nothing is wrong with your work or with the repository.** Claude's web access is a setting, and even when it is on, Claude cannot always retrieve a file by direct link. Use Path 1 instead — it is the ordinary route, not a consolation prize.

Building it in Excel, with Claude

Roughly a third of the textbook chapter is Excel choreography. Activate the Data Analysis ToolPak. Array-enter `=FREQUENCY(A2:A51,C2:C7)` with Ctrl+Shift+Enter. Delete the "More" row Excel adds. Right-click a bar, open Format Data Series, drag Gap Width to 0%. Right-click the red bars, Change Series Chart Type to Line. Set the axis maximum from 1.2 to 1.0.

That is about forty clicks per chart, and every one of them belongs to a version of Excel's user interface rather than to statistics. You will not be assessed on any of it.

The clicks were never the skill. The specification is the skill, and the specification is statistics.

Every decision the textbook labours over — how many classes, where the boundaries fall, whether they are mutually exclusive, whether all n values are accounted for, whether the bars touch, whether a pie chart is even the right chart — is exactly what you have to *tell* an assistant, and exactly what you have to *check* when it hands the workbook back.

A loose prompt and what it returns

Start here, because you will be tempted to:

> Make me a frequency distribution of this data in Excel.

Download `ch02-frequency-BROKEN.xlsx` and open it. That is what came back. It has a tidy table, a chart, and six classes. Everything in it is correct arithmetic.

Open the **Checks** sheet before you look at anything else.

Three things went wrong, and none of them is visible in the chart:

1. **Nothing is a formula.** Click cell `D5`. It is the number `5`, typed in. Change a class boundary in column B and watch the table not move.
2. **The class boundaries were never rounded.** They came straight out of $(\max - \min)/6 = 4.5$ and read "−4.20 to 0.30". Donnelly p.34 is explicit: round the estimate to a readable whole number. Nobody can hold "−4.20 to 0.30" in their head.
3. **The boundaries include both endpoints**, so a state sitting exactly on one lands in two classes. South Carolina grew 13.8% — exactly the fourth boundary — so it is counted twice. **Fifty states, fifty-one observations.** That is the textbook's rule 2, broken.

None of this required the assistant to make anything up. It answered the question it was asked. The question was underspecified.

A prompt that names every decision

PROMPT

Build me an Excel workbook from `state_hpi_ur_pop.csv`.

Sheet 1, `Data`: the raw file, headers in row 1, nothing else.

Sheet 2, `Distribution`: a frequency distribution of the `gPOP` column. Use six classes of width 5 running from −5 to 25. Columns: class label, lower bound, upper bound, frequency, relative frequency, cumulative relative frequency, plus a total row.

Classes must be **mutually exclusive**: half-open, "at least the lower bound and strictly less than the upper", so no state can fall in two classes.

Use **live Excel formulas** throughout — `COUNTIFS` referencing the bound cells, `SUM` for the total. No pasted values anywhere: I want to change a bound and watch the table and the charts update.

Sheet 3: a histogram of that distribution with **no gaps between the bars** — the data are continuous — with the axis starting at zero and both axes labelled; and an ogive of the cumulative relative frequency.

Finally, tell me which cells hold formulas and which hold constants.

Read that back and notice what it is. It is not a description of a spreadsheet. It is the whole of section 2.2: the number of classes, the class width, the boundary rule, the gap convention. **You cannot write that prompt without knowing the statistics.** That is the point, and it is why the prompt is the assessed artefact rather than the workbook.

Download `ch02-frequency-FIXED.xlsx` to see what comes back.

The ten-second demonstration

Open the fixed workbook and change **one cell**: on the `Distribution` sheet, change `C10` — the top boundary — from 25 to 22.

Idaho grew 22.8%, so Idaho now falls outside every class. The total drops to 49. The last class empties. Both charts redraw. Checks 1 and 6 turn red.

Then look at checks 2 and 3. **They still pass.** The relative frequencies are computed against the total of the classes, so they quietly renormalise to the 49 states that survived and sum to exactly 1.000. Every percentage on the sheet is now internally consistent, and describes 49 states while claiming to describe 50.

That is the Valentine's Day pie chart again, in a spreadsheet: correct arithmetic on a base nobody stated.

It is also why the first check compares against n **from the source file** rather than against the column's own total. A checklist that only asks whether a sheet agrees with itself will pass this every single time.

Set `C10` back to 25.

The checklist

Every workbook in this course carries these six checks, as live formulas rather than as advice. Run them on anything an assistant builds you — and on anything you build yourself.

#	Check	Rule it enforces
1	Frequencies sum to n, taken from the source file	Include all data values
2	Relative frequencies sum to 1.00	
3	Cumulative column ends at 1.00	
4	Frequency cells hold **formulas**, not typed numbers	Change a bound, watch it move
5	No value falls in two classes	Mutually exclusive classes
6	No empty classes	

Six checks, under a minute, and they are all rules from this chapter rather than rules about software.

Your turn

Download `ch02-frequency-STARTER.xlsx`. It has the data, the class boundaries, and the six checks already wired up. The frequency, relative frequency and cumulative columns are empty, and every check reads `-- not yet --` or `FAIL`.

Build the table until all six read `PASS`. Do it by hand, or write the prompt and have Claude do it — either is fine, and the checks do not care which. What they care about is that you can tell the difference between a workbook that is right and a workbook that merely looks right.

> **What to watch for.** These are the failures this particular task produces, in rough order of frequency:
>
> - Pasted values where you asked for formulas — by far the most common
> - Boundaries inclusive at both ends, so the frequencies sum to $n + 1$
> - The overflow class silently dropped, so they sum to less than n
> - Default bins chosen without reference to $2^k \geq n$
> - The "Other" slice quietly removed from a pie chart to make it tidier
> - A chart title that claims more than the chart shows
>
> Every one of these produces a workbook that opens cleanly and looks professional. The checklist is how you find out.

The rest of the chapter's Excel work

The same pattern covers everything else the textbook demonstrates. Each is one specified prompt and one run of the checklist:

- `COUNTIF` for a frequency distribution of **qualitative** data
- A **PivotTable** as a contingency table — pivot tables are the one Excel skill in this chapter you will use every week
- **Clustered** bar charts to compare within a category, **stacked** to compare totals across categories
- A **Pareto** chart: sorted bars plus the cumulative line on a secondary axis
- **Sparklines** — one-cell trend lines for a table of time series